

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R

PRACTICAL 14

AIM: Extracting date components using lubridate:: functions (R).

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RPROG - RStudio

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Source

```
> # Dataset: arxiv_ai.csv
> # =====
> #
> # INSTALL PACKAGES (Run Once)
> # -----
> # install.packages("lubridate")
> # install.packages("dplyr")
> # install.packages("readr")
> #
> # LOAD LIBRARIES
> # -----
> library(lubridate)

Error in library(lubridate) : there is no package called 'lubridate'

> install.packages("lubridate")

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/Purvi/AppData/Local/R/win-library/4.5'
(as 'lib' is unspecified)

also installing the dependency 'timechange'

trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/timechange_0.3.0.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/lubridate_1.9.4.zip'
package 'timechange' successfully unpacked and MD5 sums checked
package 'lubridate' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
C:/Users/Purvi/AppData/Local/Temp/RtmpIda8E2/downloaded_packages

> # =====
> # FULL R SCRIPT: Extracting Date Components using lubridate
> # Dataset: arxiv_ai.csv
> # =====
> #
> # INSTALL PACKAGES (Run Once)
> # -----
> # install.packages("lubridate")
> # install.packages("dplyr")
> # install.packages("readr")
> #
```

Environment History Connections Git Tutorial

Global Environment

df_clean	10000 obs. of 4 variables
df_part1	5000 obs. of 2 variables
df_part2	5000 obs. of 2 variables
duplicate_df	7 obs. of 3 variables
duplicates_report	2 obs. of 4 variables
job_title	3000 obs. of 18 variables
long_df	20000 obs. of 4 variables
processed_data	10000 obs. of 22 variables
unique_titles	5 obs. of 3 variables
wide_df	10000 obs. of 4 variables

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Name	Size	Modified
.gitignore	44 B	Dec 1, 2025, 11:03 AM
.RData	86.3 KB	Nov 24, 2025, 12:31 PM
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AI_Impact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
analysis.R		
prac 6.7.8.9.10		
prac 6.7.8.9&10 r prog link.R	0 B	Dec 1, 2025, 12:21 PM
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README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
arxiv_ai.csv	13.3 MB	Dec 8, 2025, 10:58 AM

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Source

```
> # 1. IMPORT DATASET AND CREATE DATE COLUMN
> # =====
> # Read CSV (readr is faster and cleaner)
> df <- read_csv("arniv_ai.csv")
Rows: 10000 Columns: 22
Column specification
Delimiter:
chr (10): authors, categories, comment, doi, entry_id, journal_ref, pdf_url, primary_category, summary, title
dtm (2): published, updated

i Use 'spec()' to retrieve the full column specification for this data.
i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
>
> # Extract only the date portion from published column
> # Example published format: "2020-01-03T18:00:02Z"
> df <- df %>%
+   mutate(
+     Published_String = substr(published, 1, 10) # Extract YYYY-MM-DD
+   )
>
> print("---- Preview of Extracted Date Strings ----")
[1] "---- Preview of Extracted Date Strings ----"
> print(head(df$Published_String))
[1] "1993-08-01" "1993-08-01" "1993-09-01" "1993-11-01" "1993-12-01"
>
> # =====
> # 2. PARSE AND EXTRACT DATE COMPONENTS USING lubridate
> # =====
> processed_data <- df %>%
+   mutate(
+     # Convert string to proper Date format
+     Actual_Date = ymd(Published_String),
+     # Extract components
+     Year_Num = year(Actual_Date),
+     Month_Num = month(Actual_Date),
+     Month_Name = month(Actual_Date, label = TRUE),
+     Day_Num = day(Actual_Date),
+     Weekday_Num = wday(Actual_Date),
+     Weekday_Name = wday(Actual_Date, label = TRUE, abbr = FALSE),
+     Quarter = quarter(Actual_Date),
+     Day_of_Year = yday(Actual_Date)
+   )
>
> print("---- Processed Data with Date Components (Top Rows) ----")
[1] "---- Processed Data with Date Components (Top Rows) ----"
> print(head(processed_data))
# A tibble: 6 x 22
```

Environment History Connections Git Tutorial

Global Environment

Object	Obs.	Vars.
df_clean	10000	4
df_part1	5000	2
df_part2	5000	2
duplicate_df	7	3
duplicates_report	2	4
job_title	3000	18
long_df	20000	4
processed_data	10000	22
unique_titles	5	3
wide_df	10000	4

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Source

```
R - R4.5.2 - ~/RPROG/
> [1] "1993-08-01" "1993-08-01" "1993-09-01" "1993-11-01" "1993-11-01" "1993-12-01"
>
> # =====
> # 2. PARSE AND EXTRACT DATE COMPONENTS USING lubridate
> # =====
>
> processed_data <- df %>%
+   # Convert string to proper Date format
+   mutate(
+     Actual_Date = ymd(Published_String),
+
+     # Extract components
+     Year_Num = year(Actual_Date),
+     Month_Num = month(Actual_Date),
+     Month_Name = month(Actual_Date, label = TRUE),
+     Day_Num = day(Actual_Date),
+     Weekday_Num = wday(Actual_Date),
+     Weekday_Name = wday(Actual_Date, label = TRUE, abbr = FALSE),
+     Quarter = quarter(Actual_Date),
+     Day_of_Year = yday(Actual_Date)
+   )
>
> print("--- Processed Data with Date Components (Top Rows) ---")
> [1] "--- Processed Data with Date Components (Top Rows) ---"
> print(head(processed_data))
# A tibble: 6 x 22
  authors categories comment doi entry_id journal_ref pdf_url primary_category published summary title
1 [anxiv. ['cs.AI']] "See h. NA http:// Journal o. http:// cs.AI 1993-08-01 00:00:00 "Secau. Dyna.
2 [anxiv. ['cs.AI']] "See h. NA http:// Journal o. http:// cs.AI 1993-08-01 00:00:00 "Marke. A Ma.
3 [anxiv. ['cs.AI']] "See h. NA http:// Journal o. http:// cs.AI 1993-09-01 00:00:00 "We de. An E.
4 [anxiv. ['cs.AI']] "See h. NA http:// Journal o. http:// cs.AI 1993-11-01 00:00:00 "As re. The
5 [anxiv. ['cs.AI']] "See h. NA http:// Journal o. http:// cs.AI 1993-11-01 00:00:00 "To su. Soft.
6 [anxiv. ['cs.AI']] "See h. NA http:// Journal o. http:// cs.AI 1993-12-01 00:00:00 "Termin. Deci.
# 11 more variables: updated <date>, Published_String <chr>, Actual_Date <date>, Year_Num <dbl>,
# Month_Num <dbl>, Month_Name <ord>, Day_Num <int>, Weekday_Num <dbl>, Weekday_Name <ord>, Quarter <int>,
# Day_of_Year <dbl>
>
> # =====
> # 3. SYSTEM DATE & TIME EXTRACTION
> # =====
>
> current_time <- now()
>
> print("--- Current System Time Details ---")
> [1] "--- Current System Time Details ---"
> print(paste("Current Year:", year(current_time)))
> [1] "Current Year: 2025"
> print(paste("Current Month:", month(current_time)))
> [1] "Current Month: 12"
```

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job_title	3000 obs. of 18 variables	
long_df	20000 obs. of 4 variables	
processed_data	10000 obs. of 22 variables	
unique_titles	5 obs. of 3 variables	
wide_df	10000 obs. of 4 variables	

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Source

```
R - R4.5.2 - ~/RPROG/
# Month_Name = month(Actual_Date, label = TRUE),
# Day_Num = day(Actual_Date),
# Weekday_Num = wday(Actual_Date),
# Weekday_Name = wday(Actual_Date, label = TRUE, abbr = FALSE),
# Quarter = quarter(Actual_Date),
# Day_of_Year = yday(Actual_Date)
#
#
# print("---- Processed Data with Date Components (Top Rows) ----")
# [1] "---- Processed Data with Date Components (Top Rows) ----"
# print(head(processed_data))
# A tibble: 6 x 22
#   authors categories comment doi entry_id journal_ref pdf_url primary_category published summary title
#   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
#1 [arxiv, ["cs.AI"]] "See h. NA http:// Journal o. http:// cs.AI 1993-08-01 00:00:00 "Sec au. Dyna.
#2 [arxiv, ["cs.AI"]] "See h. NA http:// Journal o. http:// cs.AI 1993-08-01 00:00:00 "Marke. A Ma.
#3 [arxiv, ["cs.AI"]] "See h. NA http:// Journal o. http:// cs.AI 1993-09-01 00:00:00 "We de. An E.
#4 [arxiv, ["cs.AI"]] "See h. NA http:// Journal o. http:// cs.AI 1993-11-01 00:00:00 "As re. The .
#5 [arxiv, ["cs.AI"]] "See h. NA http:// Journal o. http:// cs.AI 1993-11-01 00:00:00 "To su. Soft.
#6 [arxiv, ["cs.AI"]] "See h. NA http:// Journal o. http:// cs.AI 1993-12-01 00:00:00 "Termi. Deci.
# I'll more variables: updated <dtts>, Published_String <chr>, Actual_Date <date>, Year_Num <dbl>,
#   Month_Num <dbl>, Month_Name <chr>, Day_Num <int>, Weekday_Num <dbl>, Weekday_Name <chr>, Quarter <int>,
#   Day_of_Year <dbl>
#
#
# =====
# 3. SYSTEM DATE & TIME EXTRACTION
# =====
#
# current_time <- now()
#
# print("---- Current System Time Details ----")
# [1] "---- Current System Time Details ----"
# print(paste("Current Year:", year(current_time)))
# [1] "Current Year: 2025"
# print(paste("Current Month:", month(current_time)))
# [1] "Current Month: 12"
# print(paste("Current Days", day(current_time)))
# [1] "Current Days: 8"
# print(paste("Current Hour:", hour(current_time)))
# [1] "Current Hour: 11"
# print(paste("Current Minute:", minute(current_time)))
# [1] "Current Minute: 37"
# print(paste("Current Second:", second(current_time)))
# [1] "Current Second: 9.6288321018219"
#
#
# =====
# END OF SCRIPT
# =====
```

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